

KURVA RAMBABU

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PROFESSIONAL SUMMARY

Machine Learning and Generative AI Engineer with 4 + years of experience developing and deploying scalable AI/ML solutions across enterprise environments. Expertise in Large Language Models (LLMs), Agentic AI, Retrieval-Augmented Generation (RAG), NLP, and MLOps, with strong proficiency in Python, LangChain, LLaMA, Docker, Kubernetes, and AWS. Experienced in model fine-tuning, semantic search, vector databases, and AI-powered application development. Strong background in machine learning, deep learning, model optimization, CI/CD automation, and cloud-native architectures, delivering reliable, scalable, and cost-efficient AI solutions.

TECHNICAL SKILLS

Programming & Databases: Python, R, SQL, PostgreSQL, SQL Server, MongoDB, DynamoDB

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, XGBoost, Hugging Face Transformers, Deep Learning, NLP, Computer Vision, Predictive Modeling, Feature Engineering, Neural Networks (CNN, RNN, LSTM, Transformers)

Generative AI & LLMs: LLaMA 3, OpenAI GPT, Claude, LangChain, LangGraph, LlamaIndex, RAG, Agentic AI, AI Agents, Prompt Engineering, QLoRA Fine-Tuning, Function Calling, Tool Calling, Embeddings, Semantic Search, LLMOps

Vector Databases & Retrieval: FAISS, Pinecone, Semantic Search, Hybrid Search, Embeddings, Similarity Search, Context Re-Ranking, RAG Pipelines, Vector Indexing, Knowledge Retrieval

MLOps & Deployment: NVIDIA Triton, vLLM, TensorRT, ONNX, Docker, Kubernetes, FastAPI, Flask, GitHub Actions, CI/CD

Cloud & Infrastructure: AWS, Azure (ML Studio, AKS), Terraform, Pulumi, Prometheus, Grafana

Data Engineering: PySpark, ETL Pipelines, Feature Engineering, Data Validation, Great Expectations, Data Warehousing

Visualization & Analytics: Power BI, Tableau, Matplotlib, Seaborn, KPI Reporting, Workflow Automation, CloudWatch Dashboards

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Northern Trust | USA

Oct 2024 - Present

- Built scalable distributed ML inference infrastructure using NVIDIA Triton, vLLM, Docker, Kubernetes, AWS ECS with dynamic batching, model optimization techniques, increasing throughput by 4× and improving GPU utilization by 30%.
- Fine-tuned and deployed LLaMA-3.2 models using QLoRA, LangChain, and LangGraph to build enterprise AI assistants, reducing manual support by 40%.
- Designed and implemented enterprise-grade RAG pipelines using FAISS, Pinecone, LlamaIndex, embeddings, and semantic search techniques, improving response accuracy, contextual relevance, and reducing hallucinations by 25%.
- Developed Agentic AI workflows with multi-step reasoning, tool calling, function execution, and autonomous decision-making capabilities to automate business processes, workflow orchestration, and enterprise knowledge retrieval.
- Automated end-to-end MLOps pipelines and established observability frameworks using Terraform, GitHub Actions, CloudWatch, and Prometheus, improving deployment reliability, system monitoring, operational efficiency, and reducing cloud costs by 20%.

Machine Learning Engineer | KellTon | India

Aug 2020 - Jul 2023

- Designed, developed, and deployed machine learning regression and classification models for customer attrition prediction, improving predictive accuracy by 20% while reducing computational overhead.
- Built end-to-end machine learning pipelines using Python, Scikit-learn, Pandas, and NumPy for data preprocessing, feature engineering, model training, validation, and deployment, increasing workflow efficiency by 50%.
- Developed unsupervised learning solutions using K-Means, Gaussian Mixture Models (GMM), and Encoder-Decoder architectures to improve text embedding quality, clustering accuracy, and pattern discovery.
- Optimized machine learning and deep learning models through feature selection, hyperparameter tuning, regularization, and cross-validation techniques, improving model robustness, performance, and generalization.
- Engineered scalable ETL pipelines, automated data validation workflows using Great Expectations, and processed large-scale structured and unstructured datasets to ensure high data quality and reliability.
- Designed and administered SQL Server databases with advanced indexing, query optimization, and performance tuning techniques, significantly improving data retrieval and reporting performance.
- Leveraged AWS EC2, AWS Lambda, and cloud-based deployment frameworks to support distributed model training, scalable inference, model monitoring, and production deployment, reducing training time by 20% and improving inference latency by 25%.

EDUCATION

Master of Science in Data Science | Sep 2023 - May 2025

Saint Peter's University, Jersey City, NJ

Bachelor of Technology in Computer Science Engineering 2018 - May 2022

Malla Reddy College of Engineering and Technology, Hyderabad

CERTIFICATIONS

- Microsoft Certified: Azure AI Engineer Associate (AI-102)
- Databricks Certified Generative AI Engineer Associate